The cubic singlet two-point kernel
An all-order endpoint-orbit reduction

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Abstract

The cubic derivative singlet constructed in our preceding note obeys the q^3 probe rule at every fixed two-arc degree. We prove this by reducing arbitrary chord diagrams to 177 endpoint templates and verifying their finite polynomial identities with two independent enumerations. Every template with fewer than three bridges between the insertions cancels. The surviving three bridges give the known q-Mehler kernel, now for this particular same-coupling composite. The normalized real-time function tends to $\mathrm{sech}^6(2\mathcal{J}_s t)$ in the subsequent small-coupling limit.

1 The operator and the result

Use standard CAR and the full annealed trace $\langle Y\rangle = \mathbb{E}_J 2^{-N}\,\mathrm{Tr}\,Y$ at $\beta = \mu_{\mathrm{chem}} = 0$. Let $H = H_s$ be the Hermitian complex p-creation/p-annihilation Hamiltonian, with overlapping index sets included. Take double scaling first:

$$N, p \to \infty, \quad \lambda = p^2/N > 0, \quad q = e^{-4\lambda}, \qquad \sigma^2 = \langle H^2\rangle = \mathcal{J}_s^2/\lambda.$$

With $L = [H, \cdot]$, define

$$A = \sum_i (Lc_i)^\dagger Lc_i, \qquad T_3 = \tfrac{1}{2}\left(\sum_i c_i^\dagger L^3 c_i + \mathrm{h.c.}\right).$$

The finite-N identity $M_3 = T_3 + \frac{3}{2}LA$ removes the already known time descendant. Write $\mathcal{W}_3$ for third Gaussian *coupling* chaos, and set

$$R = \mathcal{W}_3(T_3 - a\{H, A\}/2 - bH^3), \tag{1}$$

$$a = -\frac{(1-q)^2}{1+q^2}, \qquad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}. \tag{2}$$

Its double-scaled norm is $\langle R^2\rangle = \sigma^6 V$, where

$$V = \frac{4\lambda(1+q)[1-q^3+4\lambda q^2(2q^2-q+2)]}{(1+q^2)(1+q+q^2)} > 0. \tag{3}$$

For every fixed $u, v \geq 0$ we establish

$$\boxed{\frac{\langle RH^u RH^v\rangle_{\mathrm{DS}}}{\sigma^{6+u+v}} = V P_{uv}(q, q^3).} \tag{4}$$

Here $P_{uv}(q, r)$ is the standard unit-variance probe moment: sum over Hamiltonian pairings with weight q per Hamiltonian crossing and r per crossing of the probe. Odd $u + v$ gives zero.

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2 The marked rule and a finite endpoint problem

Let K_{ij} count normal-order contractions between factors i, j of a three-factor word. The body-rank identity in the preceding note gives

$$T_3 = K_{13} - \tfrac{1}{2}(K_{12} + K_{23}), \qquad \{H, A\}/2 = \tfrac{1}{2}(K_{12} + K_{23}).$$

Thus the insertion R has marks

$$(12, 23, 13): \quad \left(-\frac{q}{1+q^2}, -\frac{q}{1+q^2}, 1\right), \qquad \text{and unmarked coefficient } -b. \tag{5}$$

Ordinary Gaussian Wick ordering deletes pairings internal to each three-slot insertion. It does not forbid a slot from pairing to an external Hamiltonian.

For a remaining pairing π, let $c(\pi)$ be its crossing number. The fixed-word Poisson-overlap rule derived from the local CAR determinant assigns to two marks the factor

$$q^{c(\pi)}(\lambda\delta + \lambda^2\alpha\beta). \tag{6}$$

Single marks carry $\lambda\alpha$ or $\lambda\beta$, and no mark carries one. The two-by-two minors give a particularly simple form. For a mark joining positions $i < j$ within one contiguous insertion,

$$\alpha_{ij} = 2 - 4\,\mathbf{1}_{\text{incident chords cross}}. \tag{7}$$

For marks (i, j) and (k, l) in the first and second insertions,

$$\delta_{ij,kl} = 4\,\mathbf{1}_{\pi(i)=l,\ \pi(j)=k}. \tag{8}$$

For example, differentiating the minor on indices $j, \pi(i)$ gives $-2\,\mathrm{sgn}(\pi(j) - \pi(i))/(s_i s_j)$, $s_i = \mathrm{sgn}(\pi(i) - i)$, which is (7). The double derivative can occur only in one minor with two marked off-diagonal entries, giving (8).

To keep the certificate in integer polynomials, define

$$D = (1+q^2)(1+q+q^2), \qquad B = 2(1-q)(1+q^3), \tag{9}$$

$$(P_{12}, P_{23}, P_{13}) = (-q(1+q+q^2), -q(1+q+q^2), D). \tag{10}$$

The total two-insertion weight, multiplied by D^2, is exactly

$$q^{c(\pi)}\left\{\lambda\sum_{i,j} P_i P_j \delta_{ij} + \lambda^2\left(\sum_i P_i\alpha_i - B\right)\left(\sum_j P_j\beta_j - B\right)\right\}. \tag{11}$$

The parameter λ is kept independent of q throughout the finite polynomial checks. This tests the two coefficients separately.

3 Exhaustion of arbitrary diagrams and cancellation

Keep the two contiguous insertion blocks and all chords incident on them. Delete every spectator chord with both endpoints outside. For any permutation of the three endpoints within either block, crossings with every spectator chord are unchanged. Their total factor can therefore be pulled out of the local permutation sum.

Let k be the number of bridges between the two blocks. There are $m = 6 - 2k$ remaining external endpoints, each colored A or B according to the block it joins, with $3 - k$ of each color. Their circular order is described by the color word and a split $s = 0, \ldots, m$ between the two exterior arcs. This classification is exhaustive regardless of the number or positions of spectator chords. Permuting slots by $S_3 \times S_3$ gives $36/k!$ distinct matchings for each template.

k	m	Templates	Matchings	Sum of (11)
0	6	$7\binom{6}{3} = 140$	5040	0
1	4	$5\binom{4}{2} = 30$	1080	0
2	2	$3\binom{2}{1} = 6$	108	0
3	0	1	6	$\lambda A_1 + \lambda^2 A_2$

The surviving polynomials are

$$A_1 = 4(1 - q^2)(1 + q^2)(1 + q + q^2)^2, \tag{12}$$

$$A_2 = 16q^2(1 + q)(1 + q^2)(1 + q + q^2)(2q^2 - q + 2). \tag{13}$$

Their sum divided by D^2 is precisely V in (3). This finite identity table is computer-assisted: the first program generates endpoint permutation orbits and factors symbolic polynomials; the second instead enumerates all perfect matchings on the reduced endpoint sets and sums integer coefficient arrays. It checks 56,106 determinant-minor values, all 176 zero templates, and both nonzero coefficient arrays independently of the orbit generator and symbolic factorizer.

For $k = 3$ every spectator Hamiltonian chord crosses either all three bridges or none, because the insertion slots are contiguous. Hence its additional factor is q^3 or one. The remaining spectator-spectator crossings contribute the usual q weights. Summing spectators gives exactly (4).

Why this is an all-order result. The bound on template size comes from the six insertion slots, not a bound on $u + v$. The proof factors out an arbitrary spectator diagram before invoking the finite table. This is distinct from checking a longer list of low-degree moments. The underlying double-scaling reduction remains a statement at each fixed degree; the real-time extension uses positive spectral measures below.

4 The full kernel and the spectral measure

Set $(z;q)_\infty = \prod_{j\ge 0}(1-zq^j)$ and use products of such factors for multiple arguments. The existing probe kernel [1, 2] gives

$$\rho_q(\theta) = \frac{(q, e^{2i\theta}, e^{-2i\theta}; q)_\infty}{2\pi}, \qquad E(\theta) = \frac{2\sigma\cos\theta}{\sqrt{1-q}}, \tag{14}$$

$$\mathcal{K}_r(\theta,\phi) = \frac{(r^2;q)_\infty}{\prod_{\epsilon,\epsilon'=\pm1}(re^{i(\epsilon\theta+\epsilon'\phi)};q)_\infty}, \qquad r = q^3. \tag{15}$$

The normalized real-time answer is

$$C_R(t) = \int_0^\pi\int_0^\pi \rho_q(\theta)\rho_q(\phi)\mathcal{K}_{q^3}(\theta,\phi)e^{it(E(\theta)-E(\phi))}\,d\theta\,d\phi. \tag{16}$$

It has $C_R(0) = 1$. The measure on the two energies is positive, absolutely continuous, and supported in $[-2\sigma/\sqrt{1-q}, 2\sigma/\sqrt{1-q}]^2$. Its pushforward to $\omega = E - E'$ is a symmetric probability measure with support $|\omega| \le 4\sigma/\sqrt{1-q}$ and no zero-frequency atom.

Justifying real time from fixed moments. At finite N introduce the normalized positive measure

$$d\nu_N(E,E') = \frac{\mathbb{E}_J 2^{-N}\,\mathrm{Tr}[R\,d\Pi_H(E)\,R\,d\Pi_H(E')]}{\langle R^2\rangle}.$$

Equation (4) supplies every limiting mixed moment. The second energy moments give tightness; bounded higher even moments give uniform integrability for each mixed monomial. Every subsequential limit therefore has the moments of the compactly supported measure in (16), which is moment determinate. Thus ν_N converges weakly to that measure, and its bounded continuous characteristic function gives (16) at every fixed real t. No finite-N exponential series has been interchanged with the limit. Portmanteau on the closed diagonal also shows that the normalized finite-N exactly static weight tends to zero.

For numerical evaluation the equivalent Jacobi expression is

$$C_R(t) = \sum_{n\ge 0} q^{3n}|\langle n|e^{-itT_q}|0\rangle|^2, \qquad (T_q)_{n-1,n} = \sigma\sqrt{\frac{1-q^n}{1-q}}. \tag{17}$$

This representation makes positivity manifest and avoids cancellation of oscillating two-energy contributions. It is the known kernel applied after the composite reduction, not a new kernel formula.

5 The small-coupling channel and computed curves

The one-matter chord representation has label $d = 3$. The fixed-degree antisymmetric-descendant contraction established in our earlier local note applies to every $d > 0$: with $h = -\log q = 4\lambda$, $C = a_L^\dagger - a_R^\dagger$, $D = C^\dagger$, and $K_\pm = (C, D)/\sqrt{h}$, its normalized limiting ladders are

$$K_+ e_n = \sqrt{(n+1)(n+2d)}\, e_{n+1}, \qquad K_- e_n = \sqrt{n(n+2d-1)}\, e_{n-1}.$$

The proof uses $[D, C] = 2(1-q^{d+n}) - (1-q)(a_L^\dagger a_L + a_R^\dagger a_R)$, the grade bound $\|C\|^2 \le h(n+1)(2d+n)$, and vanishing annihilator norms on each fixed descendant. These identities come from the established chord algebra [3].

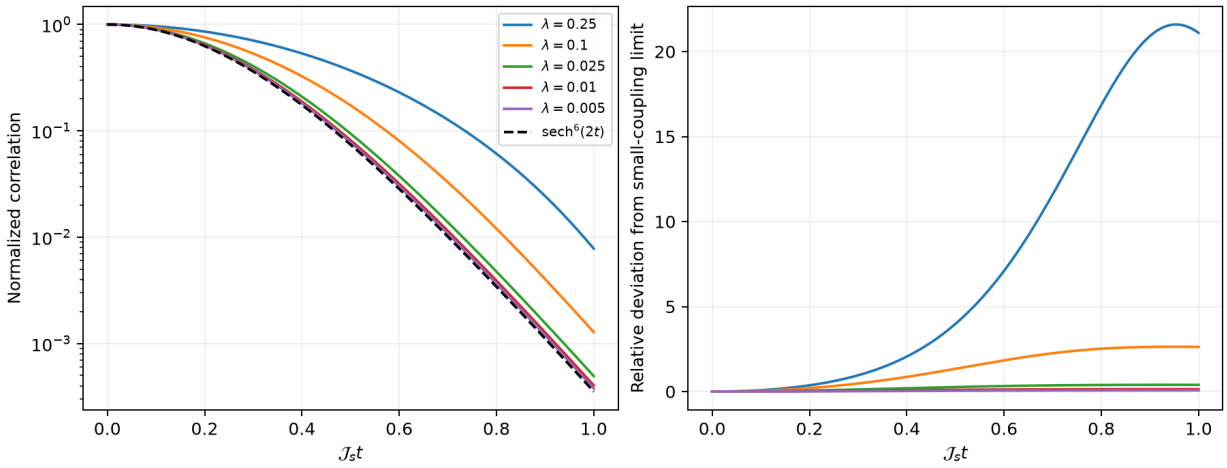
Here the physical Liouvillian is $L = 2\mathcal{J}_s(K_+ + K_-)$. At $d = 3$ its limiting Jacobi off-diagonal entries are $2\mathcal{J}_s\sqrt{(n+1)(n+6)}$. The discrete-series vacuum matrix element is therefore

$$\boxed{\lim_{\lambda \to 0^+} C_R(t) = \operatorname{sech}^6(2\mathcal{J}_s t).} \tag{18}$$

Convergence of every frequency moment, tightness, and determinacy of this exponentially decaying spectral measure justify the fixed-time limit. In the convention $C(t) = \int S(\omega) e^{it\omega}\, d\omega/(2\pi)$,

$$S_3(\omega) = \frac{\pi\omega(\omega^2 + 16\mathcal{J}_s^2)(\omega^2 + 64\mathcal{J}_s^2)}{7680\mathcal{J}_s^6 \sinh(\pi\omega/(4\mathcal{J}_s))}, \qquad S_3(0) = \frac{8}{15\mathcal{J}_s}. \tag{19}$$

This follows from the beta integral for sech^6 and is positive.



The right panel retains relative finite-λ differences. Even at $\lambda = .005$, the difference at $\mathcal{J}_s t = 1$ is about 7.09%; small absolute discrepancies at late times can otherwise be misleading. No estimate uniform in growing time is inferred.

6 Validation, sources, and the next physical question

The exact endpoint certificate covers 177 templates and 6234 distinct local matching words. Its independent integer-polynomial verifier finds the same 176 vanishing templates and surviving coefficients. The preceding stage's CAR identities and finite marked-word tests are preserved unchanged.

The kernel was evaluated for $\lambda = .25, .1, .025, .01, .005$ on $0 \le \mathcal{J}_s t \le 1$. Jacobi sizes were $33, 52, 138, 302, 574$, each compared to size $L + 16$. For $a = t_{\max}\|T_q\|$ and $L + 1 > a$, the shared first L Taylor terms imply the absolute truncation bound

$$|C_R(t) - C_R^{(L)}(t)| \le \frac{4a^L}{L!} \frac{1}{1 - a/(L+1)} < 10^{-13}. \tag{20}$$

This bounds truncation in exact arithmetic. Numerical eigenvalue roundoff is tested by refinement, not included in an interval certificate. At $\lambda = .25, .1$, independent Mehler quadratures with 128 and 192 nodes reproduce the time values and moments μ_0, μ_2, μ_4; the largest Jacobi–Mehler absolute discrepancy is below 4×10^{-14}. Full parameters, relative errors, dependencies and hashes accompany the output.

What is borrowed and what is established here. The q-Hermite recurrence, probe strength density, and Mehler kernel are established results [1, 2]. The Gaussian coupling-chaos operation defining R is not identified blindly with Hamiltonian q-Wick ordering. Hosomichi–Sato [4], Sec. 4, provide a useful precedent for proving composite relations through local chord sums, but their degenerate $d = -1/2$ fusion relation is a different construction. The new step recorded here is the all-order two-insertion reduction for the specified cubic derivative composite. We do not claim priority.

Next step. Complete the relation to the unprojected derivative family by calculating the linear Gaussian chaos of T_3. Then study mixed higher correlators of this $d = 3$ channel and the existing $d = 2$ singlet. Their mutual crossing rules and interactions have not been established by a two-insertion proof. These provide concrete new microscopic data for the Susskind program; spatial translation and mass identification still require the bulk dictionary. The full de Sitter duality remains an active research objective.

References

[1] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Sec. 7.

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[3] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725v1, Sec. 2.

[4] K. Hosomichi and M. Sato, *Degenerate chords in double-scaled SYK*, arXiv:2512.01177v2, Sec. 4, especially Eqs. (4.19)–(4.26).